

AI-DRIVEN ANALYSIS OF GLOBAL EMISSIONS: INSIGHTS AND ETHICAL LIMITS FOR SUSTAINABLE DECISION-MAKING



Burgos, J.¹, Desilva, A.¹, Wagner, J.¹, Marsellos, A.E.², Yetgin, E.¹

¹Norm Brodsky College of Business, Rider University, Lawrenceville, NJ 08648, USA

²Dept. Geology, Environment and Sustainability, Hofstra University, Hempstead, NY 11549, USA



Abstract:

Artificial Intelligence (AI) is increasingly integrated into business decision-making, particularly in environmental analysis. While AI systems can model complex relationships in environmental data, they lack the capacity to independently interpret ethical concepts such as what constitutes "sustainable" outcomes. This study investigates how AI model outputs are influenced by human-defined ethical frameworks used to classify and evaluate sustainability.

Using global datasets on greenhouse gas emissions, trade activity, and transportation, we applied K-means clustering and random forest models in RStudio to identify patterns and predict sustainability-related outcomes. K-means clustering ($k = 3$) grouped 148 countries into low ($n = 61$), intermediate ($n = 40$), and high ($n = 47$) profiles based on GDP, emissions, and mobility. A random forest regression model demonstrated moderate predictive power ($R^2 = 0.305$), identifying GDP per capita as the strongest predictor of emissions. A classification model achieved 91.67% accuracy ($k = 0.87$) in recovering clusters. An animated choropleth map indicated increasing per-capita emissions over time with shifting geographic patterns.

The findings indicate that although AI methods effectively identified patterns and improved the accessibility of environmental data, model outputs depended heavily on how sustainability was defined and labeled. This demonstrates that AI systems were not neutral tools but were shaped by the ethical assumptions embedded in their design. While AI enhances analytical capabilities, it could not determine what outcomes were ethically acceptable without human input, reinforcing the need for human oversight in sustainability decision-making.

Background

Information:

- Artificial Intelligence (AI) is increasingly integrated into business decision making, particularly in environmental analysis
- Corporations need to consider different coinciding factors when evaluating global sustainability and how it affects their operations
- With AI rapidly being used in business operations, the balance between AI-decision making and human-based decision making for business operations is an evolving point of discussion.

- This study uses global datasets on greenhouse gas emissions, trade activity measured through GDP, and transportation measured by air passenger mobility. These are used to identify patterns and predict sustainability-related outcomes

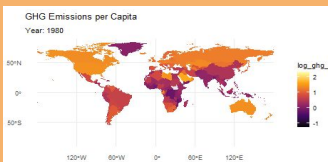


Figure 1: Animated choropleth plot visualizing global emission. Higher emissions are concentrated in more developed or industrialized countries

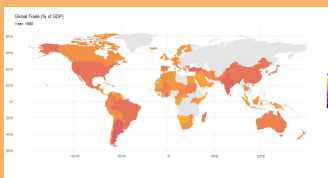


Figure 2: Animated choropleth plot visualizing global trade.

Method/Materials:

- World Development Indicators (WDI) Databank**
 - GHG, Air Passenger, and GDP data from 148 countries
 - Data from 2022 across all datasets for each country
- Natural Earth Global Country Shapefile**
 - Provided geographic boundary data and coordinate reference system
 - Spatial join key to link country data into mappable polygons
- RStudio (v.4.2.1)**
 - Extrapolated data from WDI
 - Merged datasets (GDP, GHG, and Air Passengers)
 - Packages used: ggplot2, maturalaearth, viridis, cluster, caret, randomForest, etc.
 - K-Means Clustering (elbow method, silhouette analysis)
 - Random Forest Analysis (regression and classification)

Results:

Choropleth & Animated Map Findings

- Developed regions consistently showed higher emissions and mobility.
- Developing regions showed lower values across all indicators.
- Animated map showed emissions changing over time alongside economic growth and industrialization.

K-Means Clustering Results

- Elbow and silhouette methods supported $k = 3$ as the optimal number of clusters.
- Clusters reflected low, moderate, and high development/emissions profiles.

Random Forest Regression Results

- GDP per capita was the strongest predictor of GHG emissions.
- Model explained a modest amount of variation:
 - $R^2 = 0.305$
 - MSE = 0.144

Random Forest Classification Results

- Model accurately predicted cluster membership:
 - Test Accuracy = 91.67%
 - Kappa = 0.8708
- GDP per capita had the highest variable importance.

Overall Insight

Higher economic development and mobility were generally associated with higher per-capita emissions.

Global Development, Mobility, and Emissions (2022)
Per Capita Indicators (Log Scale)

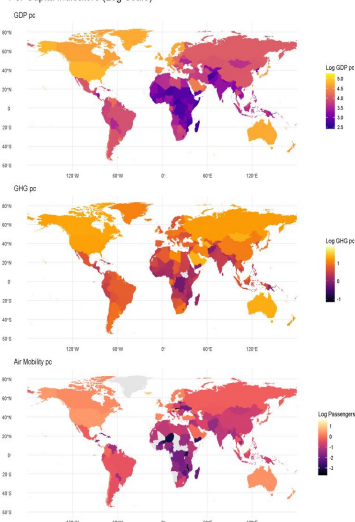


Figure 3: Initial choropleth plot visualizing global distribution of GDP, emissions, and mobility

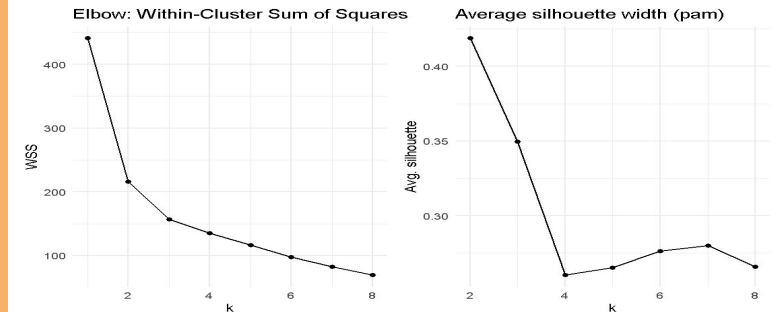


Figure 4: Elbow graph Used to determine the optimal number of clusters (k)

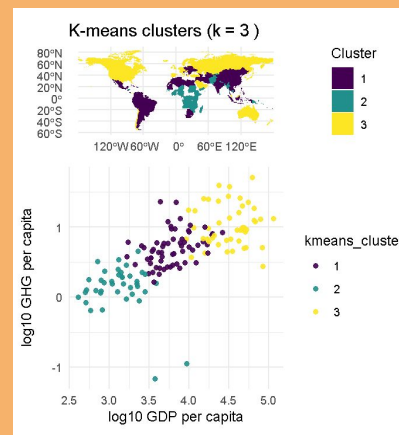


Figure 5: K-means clustering graphs visualizing the countries grouped into 3 clusters. Cluster 2 has the lowest levels across the 3 predictors, followed by Cluster 1 then Cluster 3.

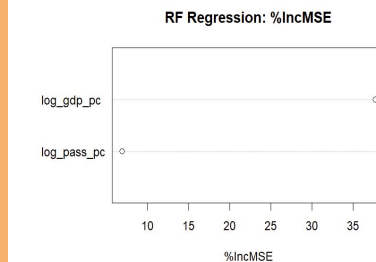


Figure 6: Random Forest Regression graph using %IncMSE to determine which variables hold the most importance in making accurate predictions.

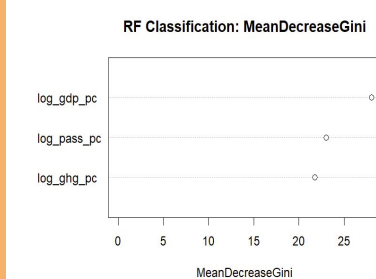


Figure 7: Random Forest Classification graph using MeanDecreaseGini. The model predicts which cluster each country belongs to using GDP, GHG, and passenger mobility as inputs

Discussion:

- AI effectively identified large-scale sustainability patterns across countries.
 - Less effective at explaining why environmental patterns exist.
- Economic development was strongly associated with higher emissions, but the relationship was not universal.
- Policy, energy systems, and efficiency likely influence environmental outcomes beyond GDP alone.
- Strong classification performance showed AI's ability to organize countries into sustainability profiles.
- Sustainability classifications still depended on human-selected variables and definitions.
- Regression results highlighted a key limitation of AI models:
 - Identifying correlations does not mean identifying causation.
- Regression model explained only 30% of emissions variation, showing limitations of purely data-driven analysis.

Conclusion:

- Human judgment remains necessary when interpreting sustainability trade-offs, ethical concerns, and business implications.
- Many sustainability factors cannot be fully captured through numerical variables alone.
- AI models are most valuable as decision-support tools rather than replacements for human decision-making.
- AI can help businesses visualize, organize, and analyze complex global sustainability data.

Acknowledgements

We would like to thank Dr. Antonio E. Marsellos & Dr. Emre Yetgin and the Norm Brodsky College of Business at Rider University for providing us with the necessary tools to conduct this research.

Credit Authorship Contribution Statement:

Burgos, J.: Editing, Rstudio coding, Figure 6, Figure 7, Abstract, Methods, Results, Discussion, Conclusion; **Desilva, A.:** Editing, Rstudio coding, Figure 4, Figure 5, Abstract, Methods, Results, Discussion; **Wagner, J.:** Editing, Rstudio coding, Figure 1, Figure 2, Figure 3, Abstract, Results, Discussion; **Marsellos, A.E.:** supervision, Rstudio coding, guidance; **Yetgin, E.:** supervision, Rstudio coding, guidance

References:

- Arel-Bundock, V. (2022). WDI: World Development Indicators and other World Bank data (R package version 2.7.8) [Computer software]. <https://cran.r-project.org/package=WDI>
- Kuhn, M. (2008). Building predictive models in R using the caret package. *Journal of Statistical Software*, 28(5), 1-28. <https://doi.org/10.18637/jss.v028.i05>
- Lian, A., & Winer, M. (2002). Classification and regression by random forest. *R News*, 2(1), 18-22. <https://cran.r-project.org/package=randomForest>
- Maechler, M., Rousseeuw, P., Struyf, A., Hubert, M., & Hornik, K. (2023). cluster: Cluster analysis basics and extensions (R package version 2.1.4) [Computer software]. <https://cran.r-project.org/package=cluster>
- Pebesma, E. (2018). Sample features for R: Standardized support for spatial vector data. *The R Journal*, 10(1), 479-488. <https://doi.org/10.18637/jss.v010.i01>
- Pedersen, T. L., & Robinson, D. (2024). gganimate: A grammar of animated graphics (R package version 1.0.9) [Computer software]. <https://gganimate.net>
- Posit Team. (2024). RStudio: Integrated development environment for R [Computer software]. Posit PBC: <https://www.rstudio.com>
- R Core Team. (2024). R: A language and environment for statistical computing [Computer software]. R Foundation for Statistical Computing. <https://www.R-project.org>
- South, A., Michael, S., & Massicotte, P. (2024). naturalEarth: World map data from Natural Earth (R package version 1.0.1) [Computer software]. <https://cran.r-project.org/package=naturalEarth>
- World Bank. (2020). Air transport, passengers carried [IS, AIR.PSGR]. World Development Indicators. <https://data.worldbank.org/indicators/IS.AIR.PSGR>
- World Bank. (2020). Total greenhouse gas emissions including LULUCF (AR CO2e) [EN.GHG.ALT.MT.CE.AB]. World Development Indicators. <https://data.worldbank.org/indicators/EN.GHG.ALT.MT.CE.AB>
- World Bank. (2020). Trade (% of GDP) [NE.TRD.GNFS.ZS]. World Development Indicators. <https://data.worldbank.org/indicators/NE.TRD.GNFS.ZS>